

Question	Answer	Marks
12(a)	Any 2 from: scattering of X-ray beam / no lead grid lack of collimation of beam / aperture large anode area large beam p.d. low / photon energy low / X-ray soft	B2
12(b)(i)	$0.81 = (e^{-1.5 \times 0.32}) / (e^{-1.5 \times x})$	C1
	$x = \mathbf{1.8 \text{ mm}}$	A1

Question	Answer	Marks
12(b)(ii)	ratio/dB = $10 \lg(0.81)$	C1
	= (–) 0.92 dB	A1

Question	Answer	Marks
12(c)(i)	probability of decay (of a nucleus)	M1